CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 5: Arithmetic Progressions

Topic-wise short notes with solved examples

An arithmetic progression is a list of numbers that goes up or down by the same amount each time. Two formulas do most of the work: one for the nth term and one for the sum of the first n terms.

1 What is an AP?

- An **arithmetic progression (AP)** is a list of numbers in which each term is obtained by adding a fixed number to the term before it.
- The fixed number is the **common difference** d. It can be positive, negative or zero.

$$d = a_2 - a_1 = a_3 - a_2 = \cdots = a_n - a_{n-1}$$

- With first term a, the general form of an AP is $a,\ a+d,\ a+2d,\ a+3d,\ \ldots$
- A **finite** AP has a last term; an **infinite** AP does not.
- To test whether a list is an AP, check that the differences between consecutive terms are all equal.

Example 1.1 Which of these are APs? (i) $4, 10, 16, 22, \ldots$ (ii) $1, 3, 9, 27, \ldots$ (iii) $5, 5, 5, 5, \ldots$

SOLUTION

(i) Differences: $6, 6, 6$. Equal, so it is an AP with $d = 6$.

(ii) Differences: $2, 6, 18$. Not equal.

(iii) Differences: $0, 0, 0$. Equal, so it is an AP with $d = 0$.

(i) and (iii) are APs; (ii) is not.

2 The nth term

- The nth term (general term) of an AP with first term a and common difference d is

$$a_n = a + (n-1)d$$

- If the AP has a last term l, the nth term **from the end** is $l - (n-1)d$.

Example 2.1 Find the 20th term of the AP $7, 11, 15, \ldots$

SOLUTION

$a = 7, d = 4, n = 20$

$a_{20} = 7 + (20-1) \times 4 = 7 + 76$

$a_{20} = 83$

Example 2.2 Which term of the AP $5, 9, 13, \ldots$ is 125?

SOLUTION

$a = 5, d = 4$. Let $a_n = 125$: $5 + (n-1) \times 4 = 125$

$(n-1) \times 4 = 120$, so $n - 1 = 30$

125 is the 31st term.

Example 2.3 Is 100 a term of the AP $3, 7, 11, \ldots$?

SOLUTION

$3 + (n-1) \times 4 = 100$ gives $n - 1 = \frac{97}{4}$, which is not a whole number.

No, 100 is not a term of this AP.

Example 2.4 The 4th term of an AP is 11 and its 9th term is 26. Find the AP.

SOLUTION

$a + 3d = 11$ and $a + 8d = 26$

Subtract: $5d = 15$, so $d = 3$ and $a = 11 - 9 = 2$

The AP is $2, 5, 8, 11, \ldots$

Example 2.5 How many three-digit numbers are divisible by 7?

SOLUTION

The numbers are $105, 112, 119, \ldots, 994$: an AP with $a = 105$, $d = 7$, $l = 994$.

$105 + (n-1) \times 7 = 994$, so $n - 1 = \frac{889}{7} = 127$

There are 128 such numbers.

Example 2.6 Find the 8th term from the end of the AP $7, 10, 13, \ldots, 184$.

SOLUTION

$l = 184, d = 3$

8th term from the end $= 184 - (8-1) \times 3 = 184 - 21$

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REMEMBER

n is a count of terms, so it must be a positive whole number. If you get a fraction or a negative value for n, the number is not a term of the AP.

3 Sum of the first n terms

- The sum of the first n terms of an AP is

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

- If the last term l is known, use the shorter form

$$S_n = \frac{n}{2}(a + l)$$

- The nth term can be found from the sums: $a_n = S_n - S_{n-1}$.
- The sum of the first n positive integers is $\frac{n(n+1)}{2}$.

Example 3.1 Find the sum of the first 20 terms of the AP $5, 8, 11, \ldots$

SOLUTION

$a = 5, d = 3, n = 20$

$S_{20} = \frac{20}{2}\left[2 \times 5 + 19 \times 3\right] = 10 \times 67$

$S_{20} = 670$

Example 3.2 Find the sum of all multiples of 6 between 1 and 200.

SOLUTION

The multiples are $6, 12, 18, \ldots, 198$, with $a = 6$, $d = 6$, $l = 198$.

Number of terms: $6 + (n-1) \times 6 = 198$ gives $n = 33$.

$S_{33} = \frac{33}{2}(6 + 198) = 33 \times 102$

The sum is 3366.

Example 3.3 How many terms of the AP $3, 7, 11, \ldots$ must be added to get a sum of 210?

SOLUTION

$\frac{n}{2}\left[6 + (n-1) \times 4\right] = 210$, so $n(2n+1) = 210$ and $2n^2 + n - 210 = 0$

$n = \frac{-1 \pm \sqrt{1 + 1680}}{4} = \frac{-1 \pm 41}{4}$, so $n = 10$ or $n = -\frac{21}{2}$

n must be a positive whole number.

10 terms.

Example 3.4 The sum of the first n terms of an AP is $S_n = 3n^2 + 5n$. Find its nth term and the common difference.

SOLUTION

$a_n = S_n - S_{n-1} = (3n^2 + 5n) - \left[3(n-1)^2 + 5(n-1)\right]$

$= 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5 = 6n + 2$

$a_1 = 8, a_2 = 14$, so $d = 6$

$a_n = 6n + 2$ **and** $d = 6$

4 Word problems

- Decide first whether the question asks for a **term** (use a_n) or a **total** (use S_n).
- Write down a, d and n before using a formula.

Example 4.1 Rekha saves ₹50 in the first week and increases her weekly saving by ₹15 every week. How much does she save in total in 12 weeks?

SOLUTION

The savings form an AP with $a = 50$, $d = 15$, $n = 12$.

$$S_{12} = \frac{12}{2}\left[2 \times 50 + 11 \times 15\right] = 6 \times 265$$

She saves ₹1590 in 12 weeks.

Example 4.2 The first term of an AP is 5, the last term is 45 and the sum of all its terms is 400. Find the number of terms and the common difference.

SOLUTION

$S_n = \frac{n}{2}(a + l)$ gives $400 = \frac{n}{2} \times 50$, so $n = 16$

$a_{16} = 45$: $5 + 15d = 45$, so $d = \frac{8}{3}$

16 terms; $d = \frac{8}{3}$

5 Useful ideas

- Three numbers in AP: take them as $a - d$, a, $a + d$. Their sum is $3a$.
- Four numbers in AP: $a - 3d$, $a - d$, $a + d$, $a + 3d$.
- a, b, c are in AP if and only if $2b = a + c$. Then b is the **arithmetic mean** of a and c.
- If every term of an AP is increased, decreased, multiplied or divided by the same non-zero number, the result is still an AP.

6 Common mistakes

- Using n instead of $n - 1$ in $a_n = a + (n - 1)d$.
- Finding d as $a_1 - a_2$. It is $a_2 - a_1$.
- Mixing up a_n (one term) with S_n (a total).
- Accepting a negative or fractional value of n.
- Forgetting that a decreasing AP has a negative d.